SPOTIFY GLOBAL MUSIC DATA ANALYSIS

Power BI Project Report

Analysis of Artist Popularity, Track Trends and Album Structures

Dataset: Spotify Global Music Dataset (2009–2025)

Source: Spotify public API data published through Kaggle

Tool: Microsoft Power BI

Prepared as an academic data analytics / visualization report

1. Executive Summary

This project presents an interactive Power BI analysis of Spotify music metadata. The dashboard is designed to examine artist popularity, track popularity, album structures, track duration, explicit-content classification, and release-year patterns. The underlying Spotify Global Music Dataset (2009–2025) was published by Warda Bilal and describes data collected from Spotify's public API for educational and research use. The dataset combines track-, artist-, and album-level information. ■cite■turn0search0■

The Power BI report uses a cleaned Spotify table containing fields such as track ID, track name, track popularity, artist name, artist popularity, artist followers, genre, album name, release date, album type, album track count and track duration. The dashboard provides filters and visual comparisons that allow a user to explore the data from different perspectives.

2. Introduction

Music streaming platforms generate large amounts of metadata that can be used to understand changes in music consumption and artist performance. Spotify provides information about tracks, artists and albums, making it useful for exploratory data analysis and visualization. The selected dataset is specifically intended for examining artist popularity, track trends and music evolution over time. ■cite■turn0search0■

The objective of this project is not to predict individual listening behavior. Instead, it focuses on descriptive analysis: identifying patterns and relationships visible in the available Spotify metadata.

3. Objectives

The main objectives of the project are:

• Analyze track and artist popularity.
• Compare explicit and non-explicit tracks.
• Examine album types and their track structures.
• Study track duration across categories.
• Explore changes in music data across release years.
• Build an interactive Power BI dashboard for easier exploration.
• Present the findings in a clear and understandable visual form.

4. Dataset Description

The public dataset is titled *Spotify Global Music Dataset (2009–2025)*. According to its Kaggle description, it contains two CSV resources: a file for modern/recent tracks and a larger file containing classic and popular tracks from 2009–2023. One public dataset view reports 8,778 rows and 15 columns for track_data_final.csv, while the separate recent-track file is much smaller. ■cite■turn0search0■turn0search1■

4.1 Main Variables

Variable	Description
track_id	Unique identifier of the track.
track_name	Name/title of the song.
track_number	Track position within its album.
track_popularity	Spotify track popularity score.
explicit	Indicates whether the track is marked as explicit.
artist_name	Name of the performing artist.
artist_popularity	Popularity score associated with the artist.
artist_followers	Number of followers associated with the artist.
artist_genres	Genre or genres associated with the artist.
album_id	Unique album identifier.
album_name	Name of the album.
album_release_date	Release date of the album.
album_total_tracks	Total number of tracks in the album.
album_type	Album, single or compilation classification.
track_duration_min	Duration of the track in minutes.

5. Data Preparation

Before visualization, the data should be checked for missing values, inconsistent text labels, duplicate records and unsuitable data types. Date fields should be interpreted as dates, popularity fields as numeric measures, and duration as a numeric measure in minutes. Categorical fields such as explicit status and album type are suitable for slicers and category comparisons.

The Power BI model represented in the uploaded report uses a cleaned Spotify data table. The dashboard design indicates that the data has been structured for aggregation and filtering rather than being presented as a raw CSV.

6. Power BI Dashboard Design

The report contains approximately 14 visual elements. These include summary indicators, category comparisons, artist-level analysis, popularity distributions, duration comparisons, and interactive filtering by release year.

Key dashboard components include:

• Total artist followers
• Total album tracks
• Artist popularity
• Explicit vs non-explicit classification

- Album type analysis
- Track popularity by album type
- Artist-level album/track analysis
- Artist popularity distribution
- Track duration by album type
- Track duration comparison for explicit and non-explicit tracks
- Release-year filtering

7. Analysis and Interpretation

7.1 Artist Popularity and Followers

Artist popularity and follower count provide two complementary indicators of artist reach. Follower count represents audience size, whereas popularity is a score used for comparative analysis. A dashboard combining both measures can help identify artists with large audiences and artists receiving strong popularity scores.

7.2 Track Popularity

Track popularity can be compared across album types and individual artists. The report's visual structure supports identification of categories containing more highly popular tracks. Popularity should be interpreted as a platform metric rather than a direct measurement of music quality.

7.3 Explicit and Non-Explicit Tracks

The explicit field divides tracks into two broad categories. Comparing duration and popularity across these groups helps determine whether the two categories show visibly different distributions. The dashboard includes explicit versus non-explicit comparisons, making this an important segmentation variable.

7.4 Album Type

Album type distinguishes structures such as albums, singles and compilations. Comparing track popularity and duration by album type can reveal whether songs released in different formats exhibit different patterns.

7.5 Track Duration

Track duration is represented in minutes. Duration can be examined by album type and explicit status. This provides a simple way to understand how song length varies between groups.

7.6 Release-Year Exploration

The report includes a year filter spanning approximately 1952–2025 in the Power BI interface. This enables users to narrow the dashboard to particular release periods and observe how the selected metrics change over time. The exact year range visible in the dashboard should be interpreted as a property of the loaded PBIX model, rather than assuming every year has equal representation.

8. Key Findings

Based on the dashboard structure and available dataset metadata, the following conclusions can be drawn from the project design:

1. Spotify music data can be effectively analyzed at track, artist and album levels.
2. Artist popularity and follower count provide useful complementary measures of artist reach.
3. Album type is a useful segmentation variable for comparing track popularity and duration.
4. Explicit classification allows the dataset to be divided into meaningful content categories.
5. Track duration can be used to compare structural differences between categories.
6. Interactive year filtering makes the dashboard suitable for exploratory time-based analysis.
7. Power BI makes it easier to combine multiple measures and categorical dimensions in one interactive report.

9. Advantages of the Dashboard

The major advantage of the Power BI implementation is interactivity. Instead of reading a static table, a user can filter the data and immediately observe how the displayed measures respond. This supports exploratory analysis and makes the project suitable for presentations and academic demonstrations.

10. Limitations

The dataset is metadata-focused and should not be treated as a complete representation of Spotify listening behavior. Popularity scores can change over time, and the number of observations is not necessarily uniform across years, artists or genres. The public dataset description also separates modern/recent tracks from older popular tracks, so users should be careful when making broad longitudinal conclusions. ■cite■turn0search0■turn0search1■

Another limitation is that the uploaded PBIX contains a compressed internal Power BI data model. This report therefore documents the dashboard and its analytical structure rather than reproducing every row of the underlying model.

11. Conclusion

The Spotify Global Music Power BI project demonstrates how structured music metadata can be transformed into an interactive analytical dashboard. By combining artist popularity, follower counts, track popularity, album information, explicit classification, duration and release-year filtering, the report provides a multi-dimensional view of Spotify music data.

The project is especially useful for demonstrating core data-analytics skills: data preparation, categorical segmentation, aggregation, KPI design, filtering and visual storytelling. The dashboard can be extended in future work with additional audio features, genre-level analysis, country-level information, time-series comparisons and predictive models.

12. Future Scope

• Add genre-level popularity analysis.
• Add advanced audio features such as danceability, energy and valence if available in the source data.
• Create year-over-year popularity trends.
• Build artist ranking and top-track pages.
• Add geographic or market-level analysis where appropriate data is available.
• Develop predictive models for popularity using historical features.
• Automate data refresh from a maintained data source.

13. References

1. Warda Bilal, *Spotify Global Music Dataset (2009–2025)*, Kaggle. Dataset description states that the data was extracted from Spotify’s public API and is intended for educational and research purposes. ■cite■turn0search0■

2. Kaggle dataset input documentation describing the two source CSV files and the 15-column track_data_final.csv structure. ■cite■turn0search1■

End of Report